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RESEARCH

Fabrication of porous ceramic buoys (Ⅰ): preparation of porous ceramic beads for buoy manufacturing

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Abstract This study presents the fabrication of porous ceramic buoys as an eco-friendly alternative to conventional expanded polystyrene (EPS) buoys, which are a major source of marine microplastics. Porous ceramic beads were synthesized using soda-lime glass waste as the base material through a foaming process involving a carbon-based blowing agent and $CaCO_3$. The optimized conditions—particle size below 25 µm, preheating at 600 °C, and foaming at 820–850 °C—produced lightweight beads with a bulk density of approximately 0.2 g cm^{-3}. These beads were consolidated with white cement to form 40 L-class ceramic buoys exhibiting a buoyancy of 15 kgf and stable floating performance for more than 60 days. The results demonstrate that porous ceramic buoys not only achieve sufficient buoyancy and mechanical integrity but also offer a sustainable solution that can fundamentally mitigate marine microplastic pollution.

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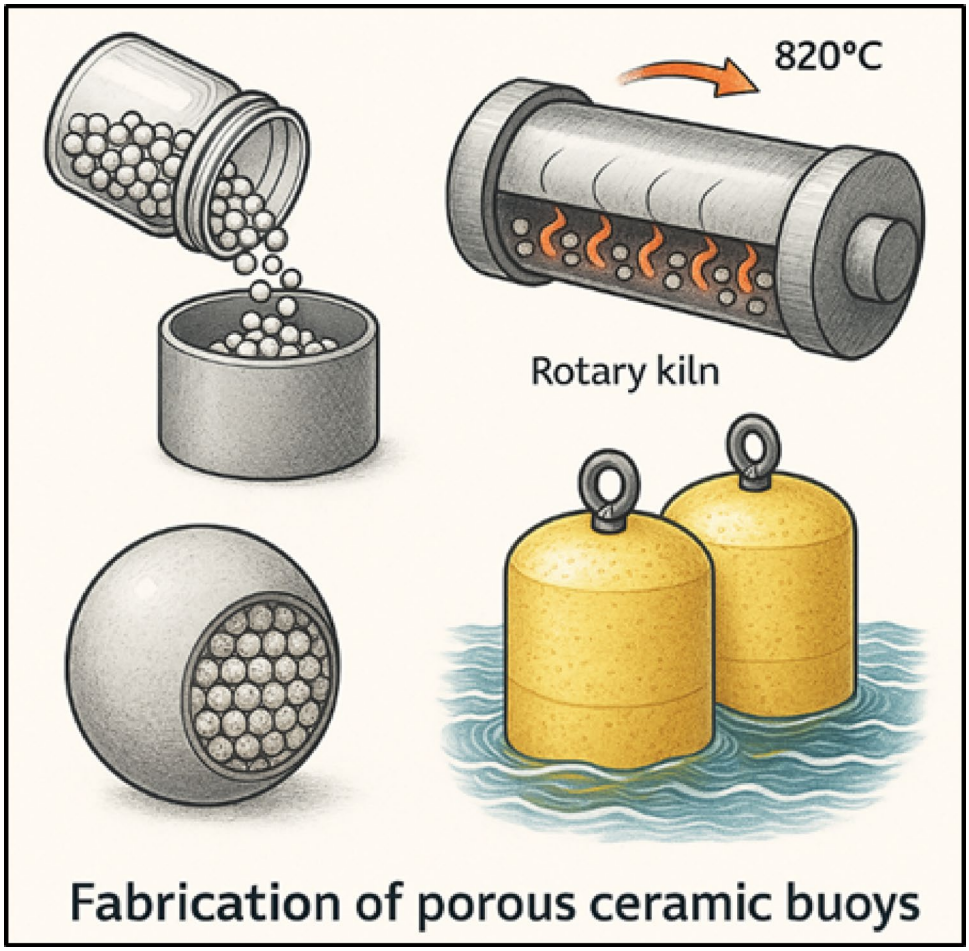
Graphical abstract



Keywords Porous ceramic buoy · Soda-lime glass · Foamed glass bead · Buoyancy · Eco-friendly materials

Introduction

Ceramic porous materials demonstrate price competitiveness by utilizing waste glass bottles destined for landfill as feedstock, reducing raw material costs. For 64 L buoys, EPS costs approximately 5,000 KRW per unit and existing eco-friendly buoys (PE-based, Al-based) average 35,000 KRW per unit, while porous ceramic bead buoys can be supplied at 20,000–25,000 KRW per unit, enabling cost savings in the fisheries sector. When considering the transition away from plastic buoys and the durability of ceramic porous materials, this approach is economically advantageous. Moreover, implementing a collection and recycling system for porous ceramic bead buoys would enable reproduction or repurposing for alternative applications (artificial sand, building insulation materials, etc.), highlighting both practical recyclability and potential expansion into other industries. Microplastics are generally defined as plastic particles smaller than 5 mm in diameter that persist in the environment without decomposition, thereby posing a serious pollution threat to the planet [1–3]. Their adverse effects are now recognized to extend from marine ecosystems to human health, which has drawn worldwide attention to their sources and mitigation strategies. Microplastics originate not only from intentionally produced products such as industrial abrasives, cosmetics, and sandblasting media but also from the degradation of larger plastic debris, synthetic textile washing, tire abrasion, and paints. Among these sources, plastic buoys used in fisheries and aquaculture have emerged as a particularly significant contributor [4].

Expanded polystyrene (EPS) buoys, which are widely employed in marine aquaculture, have caused severe environmental problems. These buoys are typically produced by expanding and molding EPS beads through steam injection, followed by adhesion and ejection from a mold. Due to their fragile structure, EPS buoys are easily fragmented

by ultraviolet radiation, waves, and wind, disintegrating into micro-sized particles that spread extensively through ocean currents. Consequently, EPS debris has been detected not only on the ocean surface but also along coastlines and in deep-sea sediments. These microplastic fragments are ingested by plankton, crustaceans, fish, and shellfish, eventually entering the food chain and causing serious marine pollution [5].

Countries engaged in coastal aquaculture have therefore made extensive efforts to address the problems associated with EPS buoys. In Korea, it is estimated that over 20 million EPS buoys are currently used in fish farms. According to a 2023 report by the Ministry of Oceans and Fisheries, more than 40% of plastic waste collected along coastlines was identified as buoy debris, and the impact continues to worsen with the expansion of offshore wind power and aquaculture industries [6]. As a countermeasure, the Korean government announced a policy in November 2023 to ban the installation of new EPS buoys in certain regions, a move that has faced opposition from fishery workers [7].

To mitigate this issue, strategies such as the retrieval and recycling of EPS buoys and the adoption of alternative materials have been implemented. Environmentally friendly buoy materials—such as expanded polyolefin (EPO), expanded polypropylene (EPP), and bio-based foams (Ecofoam)—have been introduced through government certification systems, with financial incentives provided for their use [8]. Nevertheless, although several technological attempts have been made to develop alternatives to EPS buoys [9], an innovative and fundamentally eco-friendly buoy material capable of completely eliminating microplastic generation has not yet been realized.

In this context, the present study aims to develop a porous ceramic buoy as an environmentally sustainable alternative to EPS-based buoys. Ceramic-based buoy systems have been investigated primarily for deep-sea applications due to their excellent buoyancy retention under extreme pressure [10, 11]. Inspired by the structural principle of EPS buoys—which are composed of aggregated EPS beads—the proposed ceramic buoy is fabricated by binding porous ceramic beads possessing intrinsic buoyancy. The manufacturing process of these porous ceramic beads follows the same principle as that used for the production of foamed glass bricks [12]. These beads, often referred to as *cellular glass* or *multicellular glass* beads, are inorganic materials with a glassy matrix containing independent closed pores separated by thin glass walls. The buoyancy of such beads arises directly from this unique multicellular structure.

Therefore, this study focuses on the fabrication of ultralight porous ceramic beads as the fundamental building blocks for buoy manufacturing and evaluates their feasibility for developing eco-friendly ceramic buoys that could replace EPS-based systems and fundamentally mitigate the generation of marine microplastics.

Experimental

Materials

Chemical composition of glass

Porous ceramic beads can be fabricated from various types of glassy materials. In this study, a soda–lime glass with a relatively low melting point was selected as the base material, and commercial flat glass was used as the main raw source. Table 1 lists the chemical composition of the soda–lime glass used in this work.

Binders, foaming agents, and release agents

To ensure a spherical morphology suitable for buoy fabrication, several types of binders were tested for granulation of the glass powders, including water glass (sodium silicate), polyvinyl alcohol (PVA), basalt powder, and kaolin. Carbon-based materials (petroleum coke or carbon black) and calcium carbonate ($CaCO_3$) were employed as foaming agents. During the foaming and sintering process, molten porous beads tend to adhere to each other; hence, release agents such as kaolin, calcium carbonate, or cement were introduced to prevent agglomeration. For assembling the porous ceramic beads into the final buoy structure, white Portland cement was used as the inorganic binder.

Methods

The manufacturing mechanism of the porous ceramic beads follows the same principle as that of foamed glass [12]. In this process, fine glass particles adhere near their softening temperature, encapsulating the foaming agents. The thermal decomposition or reaction between the foaming agents and glass constituents generates gases that expand within the softened glass, forming closed-cell pores. While foamed glass blocks are typically fabricated by heating a powder mixture in a tunnel kiln, the spherical porous ceramic beads in this study were produced by granulating the mixture and

Table 1 Chemical composition of Raw glasses for manufacturing porous ceramic beads

Comp	SiO_2	Na_2O	K_2O	CaO	MgO	Al_2O_3	Fe_2O_3	BaO	TiO_2
%	71.00 ±0.55	15.00 ±0.35	0.76 ±0.02	8.04 ±0.48	3.70 ±0.19	0.8 ±0.41	0.3 ±0.01	0.23 ±0.13	0.10 ±0.01

foaming it in a rotary kiln, allowing the spherical shape to be maintained throughout the process.

Preparation of glass powder

Recycled waste glass was used as received, without washing or chemical pretreatment. The collected flat glass was crushed and finely pulverized using an attrition mill to obtain powders with controlled particle sizes suitable for granulation and foaming.

Granulation of spherical glass precursors

Before foaming, the glass powders were shaped into spherical granules. To prepare these precursors, the glass powder was mixed with the selected binder and foaming agent, followed by granulation using a disk-type pelletizer into spherical particles with diameters below 5–9 mm. The green spheres were then dried. Although the disk granulation method produces a wide particle size distribution, it is highly advantageous for large-scale manufacturing. Prior to sintering, the dried granules were sieved to obtain a uniform particle size distribution for optimized foaming behavior.

Foaming and thermal stabilization

The foaming 820 °C, preheating 600 °C steps represent the key processes in producing porous ceramic beads for buoy applications [12]. A rotary kiln was used because it allows for sequential temperature control across three zones—preheating, foaming, and stabilization. The preheating zone was maintained below 600 °C, followed by the foaming zone at 700–850 °C, during which the release agent was co-introduced with the spherical precursors. After foaming, the products were stabilized at 500–550 °C and then slowly cooled to 40 °C to relieve thermal stress generated during expansion. After cooling, the release agent was removed by sieving, yielding lightweight porous ceramic beads suitable for buoy fabrication.

Fabrication of ceramic buoys

The final ceramic buoys were fabricated by combining the porous ceramic beads using an inorganic binder without damaging the internal porosity of the beads. Two fabrication routes were considered: (1) sintering-based interfacial fusion and (2) binder-assisted consolidation. In this work, the latter method was adopted for simplicity and mechanical stability. The porous beads were mixed with white cement and molded into the desired shapes using various molds. After thorough drying and demolding, solid buoy bodies were obtained, maintaining both the spherical bead structure and the inherent low density necessary for buoyancy.

Results and discussion

Preparation of porous ceramic beads

Selection and characteristics of granulation binders

The ceramic buoy was fabricated by bonding porous ceramic beads; therefore, achieving high sphericity and low density of these beads is essential. The sphericity (or sphericity index) of the porous ceramic beads is primarily determined during the granulation process, while the foaming step must maintain this spherical shape during expansion. Consequently, selecting a suitable granulation binder for shaping glass powders into spherical precursors is a critical factor influencing both morphology and final physical properties.

In previous work [13], spherical porous media for wastewater treatment were successfully fabricated using basalt, kaolin, and other inorganic binders. Based on that study, four candidate binders—PVA, basalt, kaolin, and water glass (sodium silicate)—were investigated in this research to identify the most suitable binder for preparing lightweight, highly spherical beads for buoy manufacturing. In the earlier report [13], glass powders (≤44 μm) were mixed with 0.3 wt% carbon black as the foaming agent and each binder (15–20 wt%) to produce spherical granules with diameters of 5–9 mm. These granules were preheated at 600 °C and foamed at 840 °C in a rotary kiln for 5–7 min to form porous ceramics. Following the same procedure, this study replaced the binders with 20 wt% water glass and compared the results with those from the previous binders, as shown in Fig. 1.

As illustrated in Fig. 1, the beads prepared with PVA and water glass as binders exhibited superior sphericity compared to those made with kaolin or basalt, which produced irregular or partially fused morphologies. However, the choice of binder also strongly influenced the density of the resulting porous beads. Beads synthesized using PVA or water glass showed bulk densities of 0.23–0.28 g cm^{-3}, whereas those made with kaolin or basalt exhibited higher densities of 0.31–0.35 g cm^{-3}, confirming that the binder directly affects the density of the final product. When PVA was used, it decomposed completely during foaming, leaving minimal residue, which prevented any significant density increase. In contrast, basalt and kaolin acted as binders during granulation but remained partially embedded after sintering, leading to increased overall density and making them unsuitable for buoy applications. In the case of water glass, the water content evaporated during heating, while

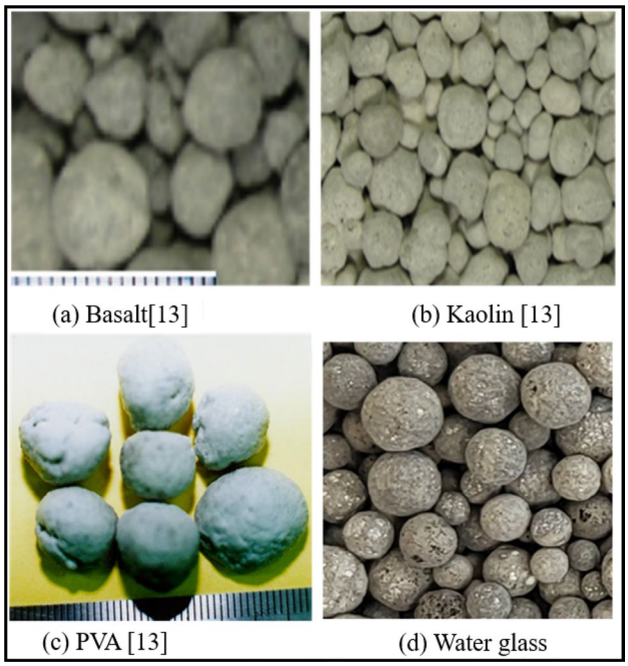


Fig. 1 Effect of the granulation binder type on the production of porous ceramic beads

the SiO_2 component remained and vitrified into the glass matrix. This residual SiO_2 could even contribute positively by reducing the final density through enhanced foaming behavior [14].

Therefore, liquid-type binders are more suitable for producing lightweight spherical porous ceramic beads for buoy fabrication. Considering the ease of handling, stability during foaming, and cost-effectiveness, water glass was determined to be the most appropriate binder for subsequent experiments.

Effects of glass particle size and foaming temperature on the density of porous ceramic beads

The most essential property of porous ceramic beads used for buoy fabrication is their low density, which directly determines the buoyancy of the final ceramic buoy. In the foaming process, the particle size of the glass powder plays a decisive role in controlling both the foaming behavior and the resulting density of the spherical porous ceramics. Generally, when the glass powder has a smaller specific surface area (i.e., larger particles), a higher sintering or foaming temperature is required to achieve sufficient melting and gas expansion. Conversely, smaller particles with a larger surface area require lower foaming temperatures and tend to produce foamed glass with lower density. According to the foamed-glass formation mechanism [12], foaming occurs when glass particles partially fuse near their softening temperature, enclosing the foaming agent. Gas

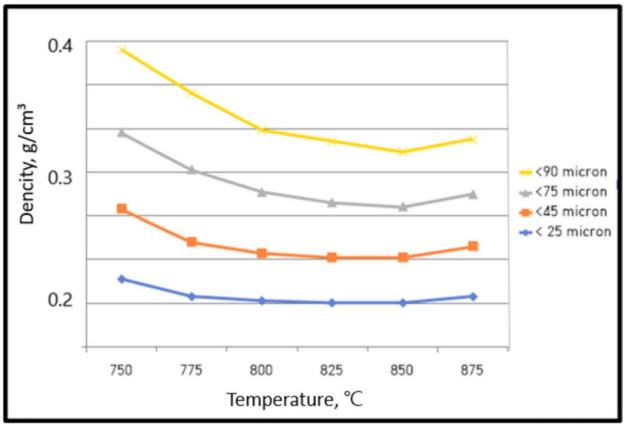


Fig. 2 Effects of foaming temperature and glass powder particle size on the density of porous ceramic beads. preheating temp; 600 °C, foaming agent 0.3 weight parts of carbon + 1.0 weight parts of $CaCO_3$

generation—either from the decomposition of the foaming agent or its reaction with glass constituents—leads to pore formation within the softened glass. The thickness of the pore walls is largely governed by the particle size of the raw glass powder; smaller glass particles produce thinner pore walls, thereby reducing the overall bulk density. Therefore, both the glass particle size and the foaming temperature are directly correlated with the density of the porous ceramic beads, which in turn determines the buoyancy of the ceramic buoy.

Figure 2 shows the measured bulk densities of porous ceramic beads produced under varying glass particle sizes (≤25 μm, ≤45 μm, ≤75 μm, and ≤90 μm) and foaming temperatures (750–875 °C). For each condition, 100 wt% glass powder was mixed with 0.3 wt% petroleum coke and 1.0 wt% $CaCO_3$ as foaming agents, and 20 wt% water glass as the granulation binder. The mixtures were shaped into spherical precursors (≤5 mm in diameter), dried, and foamed in a rotary kiln with a preheating temperature of 600 °C. After foaming, the samples were stabilized at 500 °C and slowly cooled to 40 °C. The residence time of the granules in the kiln was approximately 7–8 min. As the foaming temperature increased from 750 °C to 850 °C and the glass particle size decreased, the density of the resulting porous beads decreased significantly. The influence of particle size was particularly pronounced. When flat glass was used as the raw material and the particle size was ≤25 μm, the foamed beads exhibited a density of 0.2 g cm^{-3} under foaming conditions of 800–850 °C. For comparison, the lowest density of commercial foamed-glass blocks is about 0.15 g cm^{-3}. Given that the porous ceramic beads in this study possess an outer glassy shell, achieving a density of approximately 0.2 g cm^{-3} or lower is necessary to ensure sufficient buoyancy for practical buoy applications. Consequently, raw glass particles larger than 45 μm are considered unsuitable

for producing lightweight porous beads optimized for buoy manufacturing.

Correlation between preheating temperature and bead density

The foaming temperature has a direct and critical influence on the expansion behavior of the glass and the final density of the porous ceramic beads. During foaming, sufficient thermal energy must be supplied to allow the glass particles to partially fuse while simultaneously activating the decomposition of the foaming agents. However, if the heating time at the foaming temperature is excessive, over-foaming may occur, resulting in non-uniform pore structures or fusion between adjacent beads, which ultimately causes production defects. To ensure uniform foaming and structural integrity, a preheating stage is necessary before the foaming temperature is reached. In a rotary kiln, this can be controlled by adjusting the temperature of the inlet zone, where the spherical precursors first enter the kiln. The preheated granules then pass into the central high-temperature zone, where rapid foaming occurs within a short period of time.

Figure 3 shows the effect of preheating temperature on the density of the porous ceramic beads. Glass powders (≤45 μm) were mixed with 0.3 wt% petroleum coke and 0.5 wt% or 1.0 wt% $CaCO_3$ as foaming agents, together with 20 wt% water glass as the binder. The mixtures were granulated into spherical particles (<5 mm in diameter) and dried. The dried precursors were then introduced into a rotary kiln whose inlet temperature was varied between 250 °C and 650 °C, while maintaining the foaming zone at 830 °C. As the preheating temperature increased, the density of the resulting porous beads gradually decreased, reaching its minimum at 600 °C. Above this temperature, the density increased again, likely due to premature softening or partial coalescence of glass particles. Therefore, a preheating temperature of approximately 600 °C was identified as optimal—high enough to activate foaming reactions effectively, yet below the softening point of the glass—yielding the lowest-density porous ceramic beads.

Relationship between foaming-agent content and bead density

In the foaming process, the foaming agent must generate gas as the glass particles approach their softening point, producing internal pores. A wide range of materials—including carbonaceous substances, carbonates, and sulfates—can serve this purpose. Among these, carbon-based materials are the most widely used, while $CaCO_3$ is also known to act as an effective auxiliary foaming agent. In this study, both carbon and calcium carbonate—the typical agents

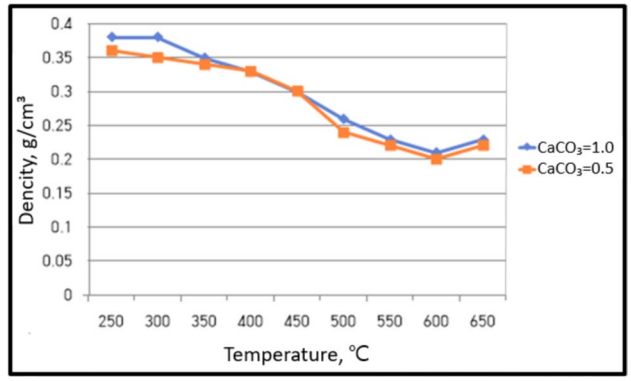


Fig. 3 Effects of preheating temperature on the density of porous ceramic beads (foaming temperature = 830 °C)

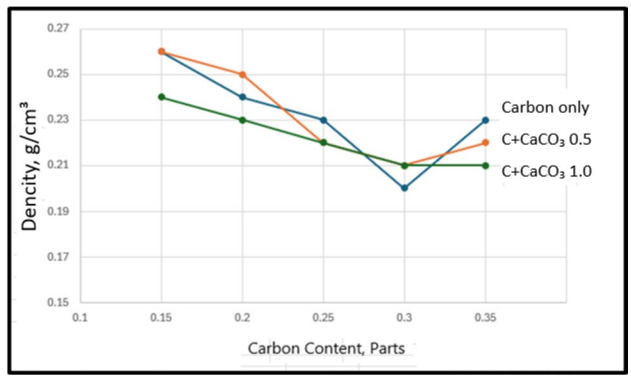


Fig. 4 Effects of foaming-agent type and content on the density of porous ceramic beads (foaming temperature = 820 °C)

used in foamed-glass block production—were employed to evaluate their combined effects on the density of the porous ceramic beads.

As shown in Fig. 4, the amount of carbon-based foaming agent was varied between 0.15 wt% and 0.35 wt%, with $CaCO_3$ added at 0.5 wt% or 1.0 wt%. The glass powders (≤45 μm) were mixed with the foaming agents and 20 wt% water glass, granulated into spheres (<5 mm), dried, preheated at 600 °C, and foamed at 820 °C. The results revealed that the lowest density was achieved when the carbon content was 0.3 wt%, regardless of the presence or absence of $CaCO_3$. Increasing the carbon content beyond 0.3 wt% led to a gradual increase in density, indicating excessive gas generation and structural collapse. This trend is consistent with observations reported for foamed-glass blocks. Interestingly, when $CaCO_3$ was co-added, the post-peak density increase was suppressed, particularly when 1.0 wt% $CaCO_3$ was used, suggesting that $CaCO_3$ can independently contribute to stable gas formation during foaming. Therefore, the optimal composition for producing low-density porous ceramic beads was determined to be 0.3 wt% carbon and 0.5 wt% $CaCO_3$ as a composite foaming agent system. Higher

additions were found to be undesirable, as they led to non-uniform pore structures and increased bulk density.

Selection of release agents and morphology of porous ceramic beads

During the foaming and sintering process in the rotary kiln, agglomeration among the ceramic beads often occurs as they melt and expand. Because the foaming temperature typically exceeds the softening or even melting point of the raw glass components, the molten surfaces of the foamed beads tend to adhere to one another, making it difficult to obtain discrete, uniform porous beads. To prevent such fusion, an appropriate release agent must be introduced together with the spherical precursors into the rotary kiln. Based on repeated experimental observations, the release agent suitable for producing porous ceramic beads should meet the following requirements:

Fig. 5 Morphology of porous ceramic beads produced using various release agents

1. It must have a melting point higher than that of the glass used.
2. It should prevent adhesion between the kiln wall and the bead surface, thereby avoiding agglomeration.
3. It should possess good flowability, allowing it to discharge simultaneously with the foamed beads.
4. It must be physically stable and chemically inert, not inducing eutectic reactions or other physicochemical interactions with the components of the glass.
5. It should be cost-effective for large-scale production.

Various metal oxides were evaluated as potential release agents, and among them, clay, kaolin, and cement were selected as suitable candidates based on their performance and availability. To verify their applicability, experiments were performed using 100 wt% glass powder with either 0.3 wt% carbon or 2.5 wt% $CaCO_3$ as the foaming agent. Each of the selected release agents—clay, kaolin, and cement—was added at 5 wt%, and the mixtures were foamed at 820–830 °C. After foaming, the beads were separated by sieving, and their morphology was examined.

As shown in Fig. 5, the resulting porous ceramic beads exhibited smooth surfaces without visible signs of sintering or fusion between particles. All samples showed a bulk density in the range of 0.19–0.21 g cm^{-3}, confirming that the selected release agents effectively prevented adhesion and maintained the spherical morphology of the beads. These results indicate that clay, kaolin, and cement are all appropriate release agents for producing porous ceramic beads under the tested conditions.

Pore structure characteristics

The internal pore structure of the porous ceramic beads was further examined to confirm their buoyancy potential. Beads were fabricated using glass powder with particle sizes ≤ 45 μm, combined with 0.3 mass% carbon and 0.5 mass% $CaCO_3$ as mixed foaming agents, and water glass as the binder. The mixture was granulated into spherical precursors (5–9 mm in diameter), dried, and subjected to preheating at 600 °C followed by foaming at 820 °C in a rotary kiln.

As shown in Fig. 6, the cross-sectional micrographs reveal that the interior of the beads consists of independent closed-cell pores, uniformly distributed throughout the structure. Such isolated pores are responsible for the buoyant behavior of the beads, as they effectively trap air and reduce the overall density. These structural characteristics confirm that the fabricated porous ceramic beads possess the necessary internal architecture to provide stable buoyancy when assembled into ceramic buoys. The closed pores formed inside the beads, as presented in Fig. 6, enable the

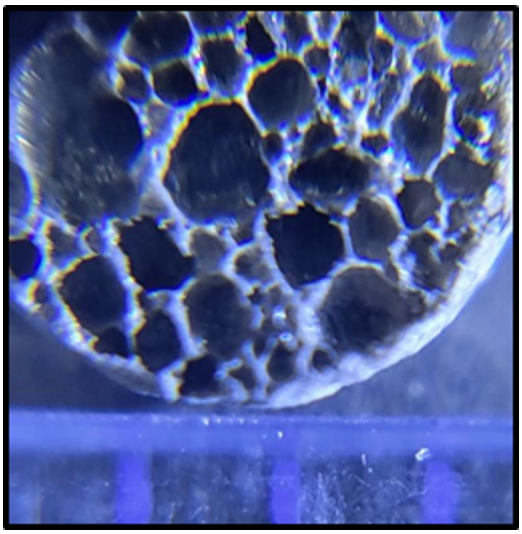


Fig. 6 Micrographs of spherical porous ceramic beads obtained at a foaming temperature of 820 °C and a preheating temperature of 600 °C, using 0.3 parts carbon and 0.5 parts $CaCO_3$ as foaming agents

Fig. 8 Buoyancy test of a shaped body manufactured by combining ceramic multicellular (porous) beads

creation of permanent buoys based on ceramic porous materials with a true density of 0.15 g/cm³ or less, as shown in Fig. 8. Unlike conventional EPS plastic buoys that gradually lose buoyancy through water infiltration during extended use or suffer instantaneous complete buoyancy loss from mechanical damage, these buoys fundamentally differ in their long-term performance characteristics.

Shaping of porous ceramic buoys based on porous beads

Porous ceramic buoys were fabricated by consolidating spherical ceramic beads with an inorganic binder. As illustrated in Figs. 7 and 100 wt% of spherical porous ceramic beads (diameter 2–5 mm, density 0.19–0.21 g cm^{-3}) were thoroughly mixed with 70–90 wt% white Portland cement as the inorganic binder. The homogeneous mixture was then packed into molds of various shapes and sizes, compacted to ensure uniform filling, and dried for approximately 9–10 h before demolding to obtain the final buoy structures. The photographs in Fig. 7 demonstrate that the spherical porous beads maintained their morphology after shaping and could be formed into a variety of configurations, confirming the high processability and structural integrity of the porous ceramic bead system.

Furthermore, as shown in Fig. 8, the shaped ceramic buoys floated stably when placed on water, and their buoyancy was maintained for more than 60 days without any noticeable degradation. This result confirms that the buoy structures, composed of interconnected porous ceramic

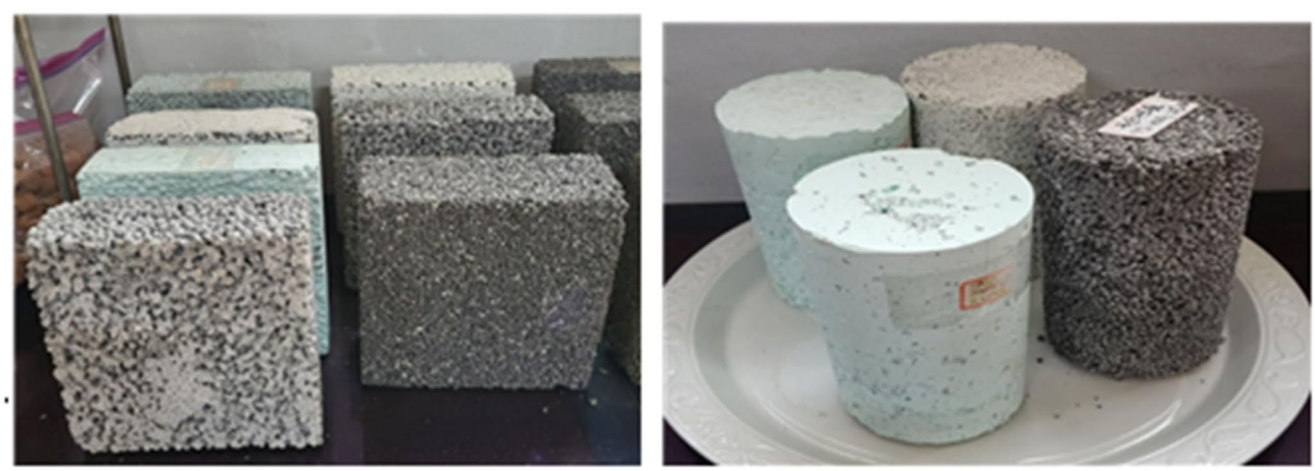


Fig. 7 Spherical porous ceramic beads prepared under foaming 820 °C, preheating 600 °C conditions of 820 °C and 600 °C, respectively

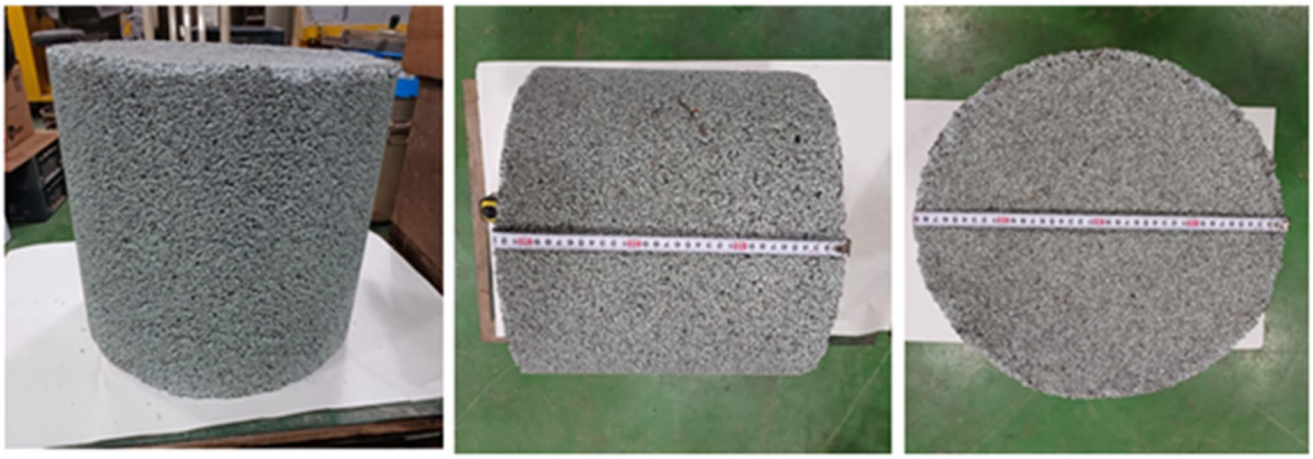


Fig. 9 40 L-class cylindrical ceramic buoys fabricated from porous ceramic beads

beads, retain sufficient strength and water resistance to serve as a durable, environmentally friendly alternative to conventional polymer-based buoys.

Performance of porous ceramic buoys

All of the fabricated porous ceramic buoys, regardless of shape, maintained stable buoyancy for over 60 days. To assess their potential for real-world application in fisheries, prototype buoys were fabricated using the porous ceramic beads developed in this study, and their performance was quantitatively evaluated according to the criteria of *eco-friendly buoy certification standards* [8]. The current eco-friendly buoy certification standard [8] in Korea was established to promote alternatives to expanded polystyrene (EPS) buoys used in aquaculture, fishing gear, and offshore facilities. It applies primarily to plastic-based buoys such as expanded polypropylene (EPP) and expanded polyethylene (EPE). Thus, directly applying these standards to ceramic materials is not entirely appropriate. Nevertheless, since the core objective of the present work is to eliminate microplastic generation through the development of environmentally benign ceramic buoys, the compatibility of our prototypes with existing certification criteria was examined as a reference framework.

Buoyancy performance of porous ceramic buoys

To evaluate the practical buoyancy performance, two 40 L-class cylindrical buoys were fabricated using porous ceramic beads with an average packing density of 0.19–0.21 g cm^{-3}, as shown in Fig. 9. White Portland cement was used as the inorganic binder. The total weights of the two buoys were 10.04 kg and 11.14 kg, respectively. The buoyancy of these 40 L-class ceramic buoys was measured using a buoyancy testing apparatus, as shown in Fig. 10. The

Fig. 10 Buoyancy measurement of a 40 L-class porous ceramic buoy

quantitative results, summarized in Table 2, indicate measured buoyancies of 16.38 kgf and 14.05 kgf, respectively, as verified by the Korea Conformity Laboratories (KCL). According to the Korean eco-friendly buoy certification standard, the required buoyancy is calculated using the formula: Certified Buoyancy = (Buoy Volume × 0.7) − Buoy Weight. For a 40 L buoy, assuming a density of 1 kg L^{-1} and a buoy weight of approximately 10 kg (with inorganic cement binder), the theoretical certified buoyancy is determined under the assumption that no water infiltrates the buoy material (i.e., a dry condition as for polymer or metal buoys).

However, in the case of the porous ceramic buoy, while the individual beads are impermeable, water can partially

Table 2 Buoyancy measurement results and analysis of 40 L-class ceramic buoys fabricated from porous ceramic beads

Samples	weight(Kg)		density (kg/m^3)	buoyancy		
	Dry Basis	Wet Basis		measured (kgf)	Certification criteria	
					Dry Basis	Wet Basis
Sample A	10.04	16.72	222.34	16.38	21.57	14.85
Sample B	11.14	16.81	275.29	14.85	17.19	11.48

Fig. 11 Impact-strength test results of a 40 L-class porous ceramic buoy

fill the inter-bead voids within the consolidated structure. Therefore, a distinction must be made between the "dry basis" weight (in air) and the "wet basis" weight (in water). Experimentally, the dry weight of the buoy was approximately 10 kg, while the wet weight increased to about 18 kg after full immersion. When applying the certified buoyancy equation, this mass difference significantly affects the calculated buoyancy. Under wet conditions, the buoyancy of the ceramic buoys exceeds the certified threshold, confirming that despite their unique structural characteristics, the porous ceramic buoys possess sufficient and stable buoyancy to meet the performance requirements for practical marine applications.

Impact resistance of porous ceramic buoys

Most eco-friendly buoys currently in use are hollow container-type structures fabricated from polymeric materials. When these buoys experience mechanical impact, the outer shell can crack or fracture, allowing water to infiltrate the interior and leading to an immediate loss of buoyancy. Consequently, the impact-resistance requirement in the eco-friendly buoy certification standard [8] specifies that the buoy surface must not exhibit cracks, fractures, or tearing when subjected to an impact load.

As shown schematically in Fig. 11, the standard impact-test method employs a 10 kg hammer shaped like a pendulum, dropped from a height of 2 m onto the surface of the buoy. The presence or absence of visible surface damage is then evaluated. The same test procedure was applied to the two 40 L-class porous ceramic buoys fabricated in this study. The test results, presented in Fig. 11, show that neither of the ceramic buoys fractured or ruptured upon impact. Only shallow surface indentations were observed at the point of contact, without any visible cracks or structural separation. This outcome demonstrates that, although localized impressions formed, the porous ceramic buoys maintained their mechanical integrity and did not lose buoyancy. The reason is that even if minor surface grooves appear, the internal porous ceramic beads are not water-permeable, and the interconnected structure effectively prevents water ingress into the buoy body. Hence, buoyancy degradation caused by impact was negligible.

In conclusion, the porous-ceramic-bead-based buoy exhibits sufficient mechanical durability and impact resistance. With minor design optimization, it is expected that such ceramic buoys could satisfy existing eco-friendly buoy certification requirements for structural strength. Furthermore, these results demonstrate the high potential of the developed porous ceramic buoy as a practical, microplastic-free alternative to conventional EPS-based buoys, contributing to the prevention of marine microplastic pollution at its source.

Conclusion

The porous ceramic beads fabricated for buoy construction exhibited a bulk density of approximately 0.2 g cm^{-3}, and the resulting ceramic buoys demonstrated strong potential as eco-friendly alternatives capable of fundamentally addressing the issue of marine microplastics.

1. The density of the porous ceramic beads directly influenced the buoyancy of the ceramic buoys. In this study, the packing density of the fabricated beads ranged between 0.19 and 0.21 g cm^{-3}.
2. The optimal conditions for preparing beads of this density were as follows:

- Raw material: soda-lime glass.
- Glass particle size: ≤ 43 μm.
- Preheating temperature: 600 °C.
- Foaming temperature: 800–850 °C in a rotary kiln.
- Foaming agents: 0.3 wt% petroleum-coke carbon and 0.5 wt% $CaCO_3$ per 100 wt% glass powder.

3. The 40 L-class ceramic buoys produced from these porous beads using white Portland cement as an inorganic binder exhibited a buoyancy of approximately 15 kgf, and the impact-resistance results further confirmed their high feasibility for practical applications.

Overall, the findings of this study demonstrate that porous-ceramic-bead-based buoys can be developed into next-generation eco-friendly buoy systems that eliminate microplastic generation at the source and offer a sustainable solution for marine environments.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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